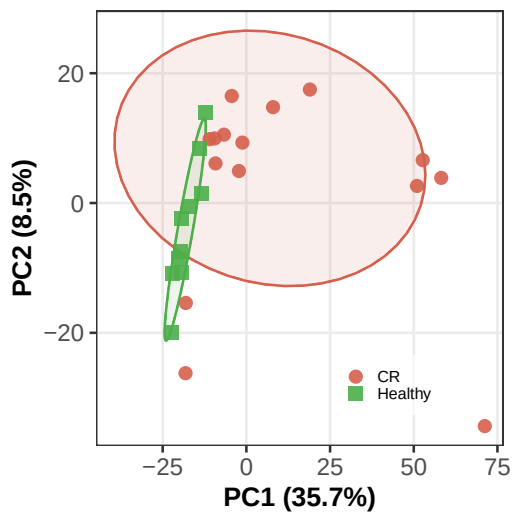


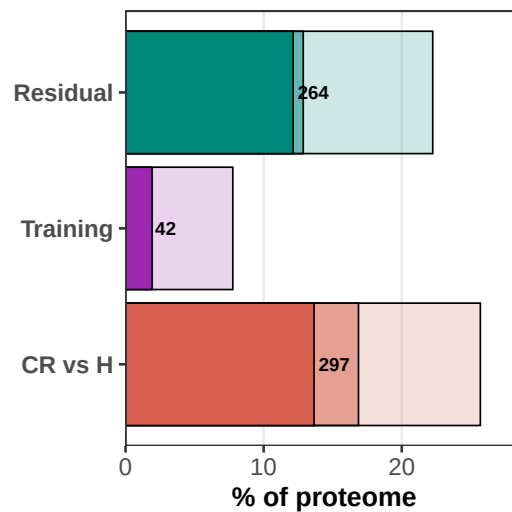
A

Sample PCA

PERMANOVA $R^2 = 0.13$, $p = 0.003$ 

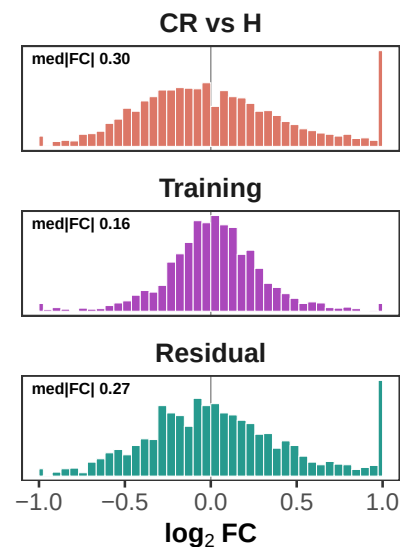
B

DEP counts

 $p / \text{FDR} / . \text{ nested per contrast}$ 

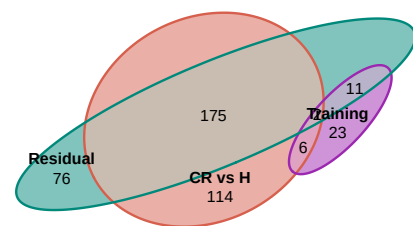
C

Effect size

 $\log_2 \text{FC per contrast}$ 

D

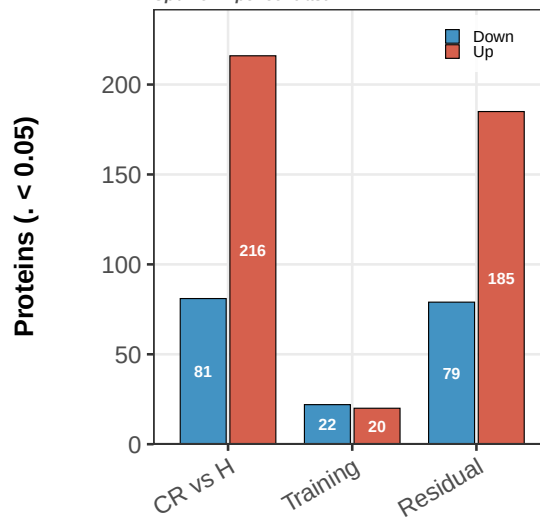
DEP overlap

 $p < 0.05$ sets

E

Direction

Up / Down per contrast



F

Pathways

fgsea $\text{FDR} < 0.05$; up / down